

AN ALTERNATE PROOF: FORBIDDING MONOCHROMATIC AND RAINBOW FOUR-CYCLES IN COMPLETE GRAPHS

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ABSTRACT. Introduced by Axenovich and Iverson [1], the mixed Ramsey spectrum, $\text{MRS}(K_n; F, H)$, is the set of numbers k such that for some k -edge coloring of K_n there is neither a monochromatic copy of $F \subseteq K_n$ nor a rainbow copy of $H \subseteq K_n$. We present an alternate proof to the finding of Axenovich and Choi [2], $\max \text{MRS}(K_n; C_4, C_4) = n$ when $n \geq 4$. It is also shown that for integers m and n , where $3 \leq m \leq n$, $\{n+2-m, \dots, n+1-m + \binom{m-1}{2}\} \subseteq \text{MRS}(K_n; C_m, C_m)$.

1. INTRODUCTION

All graphs considered in this note are finite, undirected, and simple.

A **k -edge coloring**, c , of the edge set, $E(G)$, of a graph G is a surjective mapping $c: E(G) \rightarrow [k]$ where $[k] = \{1, 2, \dots, k\}$; $c(uw)$ is the *color* of edge uw . For any subgraph $H \subseteq G$, $c[H]$ denotes the set of colors used to label the edges of H .

An edge colored graph is called **monochromatic** if all its edges have the same color. In contrast, an edge colored graph is called **rainbow** (elsewhere referred to as polychromatic or totally multicolored) if no color appears more than once.

In [1], Axenovich and Iverson apply the concept of a **mixed Ramsey spectrum**, $\text{MRS}(K_n; F, H)$, which they define as the set of numbers, k , such that for some k -edge coloring of K_n , there is neither a monochromatic copy of $F \subseteq K_n$ nor a rainbow copy of $H \subseteq K_n$ [3]. Notice that (unlike most Ramsey problems) this spectrum fixes n and finds k .

Theorem 1.1 (Axenovich and Choi [3]): *Let F be a graph that is not a star, and let H be a graph with minimum degree at least two. Then, for any natural number n , $\text{MRS}(K_n; F, H)$ is a set of consecutive integers.*

We will also use **mixed Ramsey numbers**, $\max \text{MRS}(K_n; F, H)$ and $\min \text{MRS}(K_n; F, H)$, to refer to the maximum and the minimum of $\text{MRS}(K_n; F, H)$. These numbers are closely related to some more familiar types of Ramsey numbers [2].

Assuming $\text{MRS}(K_n; F, H)$ is nonempty, the value of $\max \text{MRS}(K_n; F, H)$ is less than or equal to the **anti-Ramsey number**, $\text{AR}(n; H)$, which is the maximum number of colors, k , such that there exists a k -edge coloring of K_n with no rainbow copy of $H \subseteq K_n$. Secondly, the value of $\min \text{MRS}(K_n; F, H)$ is related to the **multicolor Ramsey number**, $R_k(F)$, defined as the smallest n such that for every k -edge coloring of K_n there is a monochromatic copy of $F \subseteq K_n$.

In a previous paper [5], we found $\text{MRS}(K_n; K_3, K_3)$.

Theorem 1.2 (DeMars and Johnson [5]): *For all $n \geq 3$, $\text{MRS}(K_n; K_3, K_3) = \{g(n), \dots, n-1\}$ where $g(n) \in \{\lceil 2 \log_5 n \rceil, \lceil 2 \log_5 n \rceil + 1\}$.*

Key words and phrases. mixed-ramsey; edge coloring; rainbow; monochromatic; four-cycle.

It should be noted that the value of $\min \text{MRS}(K_n; K_3, K_3)$ was obtained from results in Chung and Graham's classic work [4].

Theorem 1.3 (Chung and Graham [4]): *Let $f(k)$ be the largest value of n such that it is possible to edge color K_n , with k or fewer colors, so that every copy of $K_3 \subseteq K_n$ is neither monochromatic nor rainbow. Then*

$$f(k) = \begin{cases} 5^{\frac{k}{2}} & \text{if } k \text{ is even} \\ 2 \cdot 5^{\frac{k-1}{2}} & \text{if } k \text{ is odd} \end{cases}$$

This led us to inquire about $\text{MRS}(K_n; C_4, C_4)$ and more generally $\text{MRS}(K_n; C_m, C_m)$. In [2] Axenovich and Choi showed the following result. Let $s(G) = \min\{s : G \subseteq K_{s,r}, s \leq r \text{ for some } r\}$.

Theorem 1.4 (Axenovich and Choi [2]): *For integer $n \geq 3$, any graph G that is not a star, the following holds:*

$$\max \text{MRS}(K_n; G, C_4) = \begin{cases} \frac{4}{3}n + O(1) & \text{if } \chi(G) = 2 \text{ and } s(G) \geq 4, \text{ or } \chi(G) \geq 3, \\ n & \text{otherwise.} \end{cases}$$

Theorem 1.5 (Axenovich and Choi [2]): *For integer $m \geq 3$, any graph bipartite G that is not a star, there exist constants n_0 and g such that for all $n \geq n_0$, the following holds:*

$$\max \text{MRS}(K_n; G, C_m) = \begin{cases} n \left(\frac{m-2}{2} + \frac{1}{m-1} \right) + O(1) & \text{if } \chi(G) = 2 \text{ and } s(G) \geq 4, \text{ or } \chi(G) \geq 3, \\ n \left(\frac{s(G)-2}{2} + \frac{1}{s(G)-1} \right) + g & \text{otherwise.} \end{cases}$$

This is where g is the Erdős-Rado number, which is determined by G and m . When we let $G = C_4$ in Theorem 1.4 we are left with the Corollary below.

Corollary 1.6 (Axenovich and Choi [2]): *For every integer $n \geq 4$,*

$$\max \text{MRS}(K_n; C_4, C_4) = n.$$

The research of Axenovich and Choi is undoubtedly noteworthy, as their work presents groundbreaking findings that may have implications extending far beyond the scope of our current manuscript. However, it is important to note that our primary focus lies in presenting an alternate proof of Corollary 1.6 that is both accessible and understandable to a broad audience.

To this end, our proof is designed to be more elementary, providing a clear and concise presentation of the key concepts and ideas that underlie our arguments. In doing so, we aim to not only establish the validity of our claims but also to facilitate a deeper understanding of the underlying principles that govern our analysis.

Moreover, our proof is carefully structured to highlight the intermediate results that form its backbone. These results, which are of independent interest and may have potential applications in other fields of study, are presented in a manner that allows for their isolation and further analysis.

While the broader implications of Axenovich and Choi's research are certainly worth noting, it is important to recognize that our manuscript is not concerned with these broader implications. Rather, our focus is on providing a self-contained proof that is both rigorous and accessible, and that offers insights into the fundamental principles that underlie our analysis.

We also prove the following Theorem.

Theorem 1.7: For integers m and n , where $3 \leq m \leq n$,

$$\left\{ n + 2 - m, \dots, n + 1 - m + \binom{m-1}{2} \right\} \subseteq \text{MRS}(K_n; C_m, C_m)$$

2. PROOF OF THEOREM 1.7 AND RELATED COROLLARIES

While Theorem 1.5 gives for an odd integer m , $\max \text{MRS}(K_n; G, C_m) = n \left(\frac{m-2}{2} + \frac{1}{m-1} \right) + O(1)$, Theorem 1.7 gives us $\max \text{MRS}(K_n; G, C_m) \geq n + 1 - m + \binom{m-1}{2}$ and a subset of the spectrum for $\text{MRS}(K_n; C_m, C_m)$. Our proof of Theorem 1.7 is designed to be elementary, providing a clear and concise presentation of key concepts and ideas. We aim to establish the validity of our claims while promoting a deeper understanding of the underlying principles of the spectrum.

Before showing the proof of Theorem 1.7, we will first define a join. The **join**, $G \vee H$, of two vertex-disjoint graphs G and H has $V(G \vee H) = V(G) \cup V(H)$ and $E(G \vee H) = E(G) \cup E(H) \cup \{uv : u \in V(G), v \in V(H)\}$.

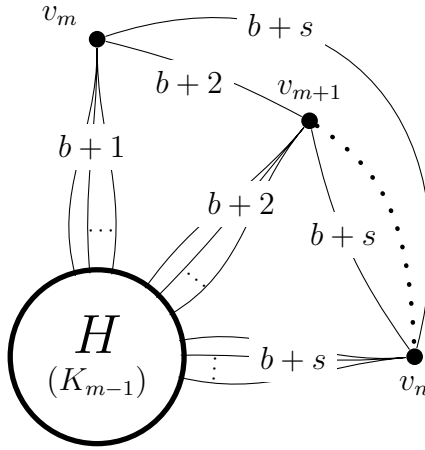


FIGURE 1. $K_{m-1} \vee v_m \vee \dots \vee v_n$; $s = n + 1 - m$

Proof of Theorem 1.7. Let $m, n \in \mathbb{Z}$ such that $3 \leq m \leq n$. Consider $K_{m-1} \subseteq K_n$; we will refer to this subgraph, K_{m-1} , as our base, H . In the following steps, we will construct an edge coloring on K_n forbidding monochromatic and rainbow C_m 's. First color the edges of H with color set B , where $B = \{1, \dots, b\}$ and $1 \leq b \leq \binom{m-1}{2}$ with all b colors appearing. Secondly, we will color all the edges joining H to v_m with color $b + 1$. Note that $V(K_n) \setminus V(H) = \{v_m, \dots, v_n\}$. In the coloring thus far, the complete graph $K_{m-1} \vee v_m$ contains neither monochromatic nor rainbow C_m 's.

Now concerning additional edges incident to vertices $v_{m+1}, \dots, v_n$. Color edges joining $K_{m-1} \vee v_m$ to v_{m+1} with color $b + 2$. Continue in this way: color the edges from $V(H) \cup \{v_m, \dots, v_{m+t}\}$ to v_{m+t+1} , $1 \leq t \leq n - m - 1$, with color $b + t + 2$. Notice, the value of $|c[K_n]|$ ranges from

- (1) $b + n + 1 - m = n + 2 - m$, when $b = 1$, to
- (2) $b + n + 1 - m = n + \binom{m-1}{2} + 1 - m$, when $b = \binom{m-1}{2}$.

To see that K_n , with any of these edge colorings, contains neither monochromatic nor rainbow m -cycles, consider an arbitrary $C \simeq C_m$ in K_n . Since $H \vee v_m$ contains neither monochromatic nor rainbow C_m 's, we may as well assume that $V(C)$ contains at least one vertex v_j , $m < j \leq n$.

Let $\alpha \in \{m+1, \dots, n\}$ be the largest integer such that $v_\alpha \in V(C)$. By the way the coloring c is defined, the two edges of C incident to v_α bear the same color, so C is not rainbow. Now, because $m = |V(C)| \geq 3$, C has an edge uw , $u, w \in V(H) \cup \{v_m, \dots, v_{\alpha-1}\}$; again, by appeal to the way the coloring is defined, the color on uw is different from the color shared by the two edges incident to v_α , so C is not monochromatic. $\square$

Proposition 2.1 (Axenovich and Choi [3]): *If a graph F is not a star, a graph H has minimum degree at least two, and $k \in \text{MRS}(K_n; F, H)$, then $k+1 \in \text{MRS}(K_{n+1}; F, H)$.*

Proof. Let K_n be k -edge colored with all colors of $[k]$ appearing, with no monochromatic or rainbow $C_m \subseteq K_n$. Now join K_n to $v_{n+1} \in V(K_{n+1}) \setminus V(K_n)$ and color the new joining edges with $k+1$. With respect to the edge coloring, K_{n+1} clearly contains no monochromatic or rainbow $C_m \subseteq K_{n+1}$. Therefore $k+1 \in \text{MRS}(K_{n+1}; C_m, C_m)$. $\square$

Corollary 2.2: *For integer n , where $n \geq 4$,*

$$\{n-2, n-1, n\} \subseteq \text{MRS}(K_n; C_4, C_4).$$

Proof. This result was obtained from Corollary 1.6 and Theorem 1.7. $\square$

3. PRELIMINARY LEMMAS FOR COROLLARY 1.6

Fact 3.1: *If G is connected, C is a cycle in G , and $e \in E(C)$, then $G - e$ is connected.*

Fact 3.2: *If G is connected and acyclic then $|V(T)| - 1 = |E(T)|$.*

Corollary 3.3: *If G is connected and $|V(G)| \leq |E(G)|$ then G contains a cycle subgraph.*

Corollary 3.4: *If G is a connected graph and if*

- (1) $|V(G)| + 1 \leq |E(G)|$, then G contains at least two different cycle subgraphs, C_1 and C_2 . Furthermore,
- (2) if $|V(G)| + 1 = |E(G)|$, then G satisfies the following: If e_1 is an edge on some cycle subgraph of $C_1 \subseteq G$, and e_2 is an edge (where $e_1 \neq e_2$) on some other cycle subgraph $C_2 \subseteq G - e_1$, then $G - (e_1 \cup e_2)$ is a tree.

Erdős, Rubin, and Taylor [6] use the concept of the **generalized Θ graph**, denoted $\Theta_{r,s,t}$. Define $\Theta_{r,s,t}$ as “the graph consisting of two end vertices u and w meeting three internally vertex disjoint paths containing r , s , and t edges, respectively.”

Corollary 3.5: *If G is connected and $|V(G)| + 1 = |E(G)|$, then either*

- (1) G contains two edge-disjoint cycles, or
- (2) G contains exactly three cycles, and together they form a Θ graph.

Lemma 3.6: *Let K_n be edge colored with at least $n+1$ colors appearing. Then K_n has a rainbow connected subgraph G such that $|V(G)| + 1 = |E(G)|$.*

Proof. Choose $n+1$ edges of K_n bearing $n+1$ distinct colors. Let H be the subgraph of K_n induced by these edges. Notice that H is rainbow. Then $n+1 = |E(H)| = |V(K_n)| + 1 \geq |V(H)| + 1$.

Since $|E(H)| > |V(H)|$, some component H' of H , satisfies $|E(H')| > |V(H')|$. If $|E(H')| = |V(H')| + 1$, take $G = H'$. If $|E(H')| > |V(H')| + 1$, obtain G by removing edges from cycle subgraphs of H' (with each cycle after the first being a subgraph of the graph arrived at by the previous removal) to obtain a connected subgraph G such that $|E(G)| = |V(H')| + 1 = |V(G)| + 1$. Since H is rainbow, its subgraph G is rainbow. $\square$

Proposition 3.7: *Let K_n be edge colored with at least $n + 1$ colors appearing. Then K_n has a rainbow connected subgraph H bearing at least $|V(H)| + 1$ colors where either*

- (1) H contains two edge-disjoint cycles, or
- (2) H contains exactly three cycles and they form a Θ graph.

Lemma 3.8 (Gyárfás, Sárközy, Sebő, and Selkow [7]): *Suppose that K_n is edge colored, and for some integer m , there is a rainbow C_m subgraph of K_n . Then there is a rainbow $C_3 \subseteq K_n$.*

Proof. We can assume that $m > 3$. Let $e = uv$ be a chord of the rainbow C_m . There are two edge-disjoint paths, P and Q , on C_m with end vertices u and v . Because C_m is rainbow, the color of e in the edge coloring of K_n appears on at most one of P, Q . Therefore, either $P \cup e$ or $Q \cup e$ is a rainbow cycle of order $s < m$. If $s = 3$, we are done. Otherwise, we repeat the process in the argument preceding until we obtain a rainbow C_3 . $\square$

Lemma 3.9: $k \notin \text{MRS}(K_4; C_4, C_4)$ for all $k \geq 5$.

Proof. Let K_n be k -edge colored where $k \geq 5$. $|E(K_4)| = 6$, so exactly one color is repeated exactly once if $k = 5$. Remove one edge, e , associated with the repeated color. Then $K_4 - e$ is rainbow. $K_4 - e$ contains a C_4 , which is necessarily a rainbow C_4 . $\square$

We define a **B_t graph** as the graph consisting of two edge disjoint triangles, joined by path of length $t - 1$.

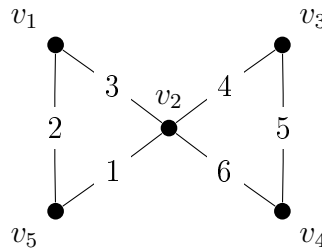


FIGURE 2. A rainbow subgraph, $B_1 \subseteq K_5$.

Lemma 3.10: *If $n \geq 5$, K_n is edge colored, and $B_1 \subseteq K_n$ is rainbow, as depicted in Fig. 2, then $B_1 \cup v_1v_3$ contains a rainbow C_4 as a subgraph.*

Proof. Referring to Fig. 2, if we forbid rainbow $C_4 \subseteq B_1 \cup v_1v_3$, we then we have $c(v_1v_3) \in \{3, 5, 6\} \cap \{1, 2, 4\} = \emptyset$. Therefore, there is a rainbow $C_4 \subseteq B_1 \cup v_1v_3$. $\square$

Corollary 3.11: $k \notin \text{MRS}(K_5; C_4, C_4)$ for all $k \geq 6$.

Proof. Let K_n be k -edge colored where $k \geq 6$ and all k colors appearing. By Proposition 3.7, we know there is a connected rainbow $H \subseteq K_5$ with $|V(H)| + 1$ colors appearing on H

containing at least two cycles. Assume there are no C_4 's in H . Since $|V(H)| + 1 \leq 6$, either H contains a C_5 or H contains two C_3 's.

Case 1. If H contains a C_5 , then any edge in H bearing a sixth color (not on the C_5), must be a chord across C_5 , so H contains a rainbow C_4 .

Case 2. There are two C_3 's sharing either an edge or a vertex.

Case i. If the two C_3 's share an edge, then H contains a C_4 , which is rainbow because H is rainbow.

Case ii. If the two rainbow C_3 's share exactly one vertex, refer to Lemma 3.10. □

Corollary 3.12: *For $n \geq 5$, if we have K_n edge colored with any number of colors appearing, with no rainbow C_4 's, then we can conclude that there are no rainbow B_1 subgraphs in K_n .*

Corollary 3.13: *$n + i \notin \text{MRS}(K_n; C_4, C_4)$ where $n \in \{4, 5\}$ and $i \geq 1$.*

Proof. This claim combines those Lemma 3.9 and Corollary 3.11. □

Given a nonempty set $S \subseteq V(G)$, the subgraph $G[S] \subseteq G$ is said to be **induced** by S if: $u, w \in S$ are adjacent in $G[S]$ if and only if u and w are adjacent in G .

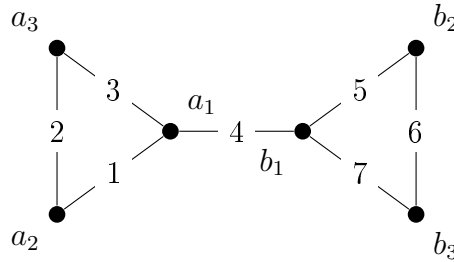


FIGURE 3. A rainbow subgraph, $B_2 \subseteq K_n$.

Let B_2 be such a rainbow copy described as well as depicted in Fig. 3.

Lemma 3.14: *Let $k \geq 7$, $n \geq 6$, and suppose that there is an edge coloring of K_n such that there are no rainbow copies of $C_4 \subseteq K_n$, but there is a rainbow copy of $B_2 \subseteq K_n$.*

Then every edge in $E(K_n[V(B_2)]) \setminus E(B_2)$ must be colored four (the color of edge a_1b_1). Consequently, there exists a monochromatic copy of C_4 in K_n , under this coloring.

Proof. Let c denote the coloring of $E(K_n)$. Since there are no rainbow C_4 's in K_n , $c(a_3b_2) \in \{3, 4, 5\} \cap \{2, 6, c(a_2b_3)\}$, and similarly $c(a_2b_3) \in \{1, 4, 7\} \cap \{2, 6, c(a_3b_2)\}$. It now follows that $c(a_3b_2) = c(a_2b_3) = 4$. Then $c(a_3b_1) \in \{1, 2, 4\} \cap \{6, 7, 4\} = \{4\}$. Given the symmetry of the situation, it follows that $c(a_3b_3) = c(a_1b_2) = c(a_1b_3) = c(a_2b_1) = c(a_2b_2) = 4$. So there exists a monochromatic $a_3b_2a_2b_3a_3 = C_4 \subseteq K_6$. □

Lemma 3.15: *Assume $n > 6$. Color the edges of K_n . Let $B_3 \subseteq K_n$. Then K_n contains a monochromatic or rainbow C_4 .*

Proof. Let $|V(P)| = 3$ as depicted in Fig. 4, then we either have at least a rainbow $B_1 \subseteq K_n$ or a rainbow $B_2 \subseteq K_n$.

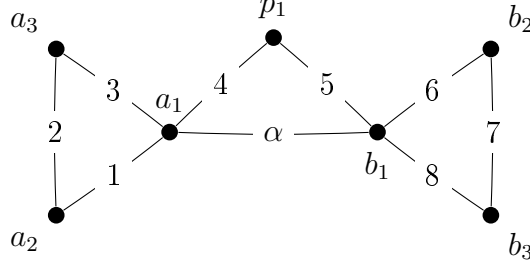


FIGURE 4. A rainbow subgraph $B_3 \subseteq K_n$ with an additional edge, a_1b_1 , where $c(a_1b_1) = \alpha$.

- Case 1.* If $\alpha \in \{1, 2, 3, 6, 7, 8\}$ then $a_1p_1b_1$ is rainbow, and either $a_1a_2a_3 \cup a_1p_1b_1$ is rainbow or $b_1b_2b_3 \cup a_1p_1b_1$ is rainbow. In both cases we have a rainbow $B_1 \subseteq K_n$ and should refer to Lemma 3.10.
- Case 2.* If $\alpha \notin \{1, 2, 3, 6, 7, 8\}$ then we have a rainbow $B_2 \subseteq K_n$ and should refer to Lemma 3.14.

□

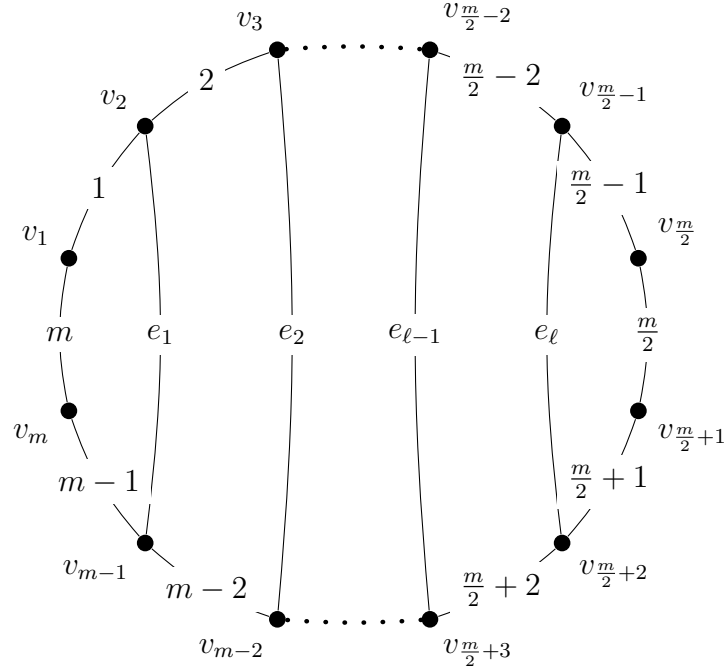


FIGURE 5. A rainbow $C_m \subseteq K_n$ (where m is even) with strategically placed chords, e_i , for integers $1 \leq i \leq \ell$.

Lemma 3.16: Assume $n \geq m \geq 6$ and m is even. Let K_n be edge colored and assume that $C_m \subseteq K_n$ is rainbow. Then there is a rainbow $C_4 \subseteq K_n$.

Proof. Let $C_m \subseteq K_n$ be m -edge colored with colors one through m , as depicted in Fig. 5. Suppose there is not a rainbow $C_4 \subseteq K_n$. Drop a chord, e_1 , across C_m from v_2 to v_{m-1}

as depicted in Fig. 5. Since there is not a rainbow $C_4 \subseteq K_m$, $c(e_1) \in \{1, m-1, m\}$. If $m \geq 6$, draw another chord, e_2 , from v_3 to v_{m-2} , then $c(e_2) \in \{c(e_1), 2, m-2\}$. If this process continues, there are $\frac{m-2}{2} + 1 = \ell$ chords drawn across C_m . So continue this process until a chord, e_ℓ , is drawn from $v_{\frac{m}{2}-1}$ to $v_{\frac{m}{2}+2}$. Because $c(e_\ell) \in \{c(e_{\ell-1}), \frac{m}{2}-2, \frac{m}{2}+2\} \cap \{\frac{m}{2}-1, \frac{m}{2}, \frac{m}{2}+1\}$ it follows that $c(e_\ell) = c(e_{\ell-1}) = \frac{m}{2}$. By a similar argument $c(e_{\ell-1}) = c(e_{\ell-2}) = \dots = \frac{m}{2}$. However, since $m \geq 6$, the C_4 on vertices $v_1 v_2 v_{m-1} v_m$ is rainbow. $\square$

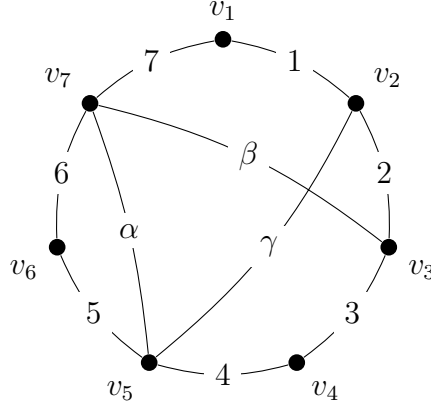


FIGURE 6. A rainbow $C_7 \subseteq K_n$ where $n \geq 7$ with chords colored α , β , and γ .

Lemma 3.17: *Let $k, n \geq 7$ be integers and K_n is k -edge colored. If there is a rainbow $C_7 \subseteq K_n$, then there is a monochromatic $C_4 \subseteq K_n$.*

Proof. Let $k, n \geq 7$ be integers. Suppose that K_n is k -edge colored and there is a rainbow C_7 as depicted in Fig. 6. Label the chords with colors α, β , and γ as depicted in Fig. 6.

We will examine the color of several edges of C_4 and assume that no C_4 is rainbow. Consider the cycles $v_1 v_2 v_3 v_7$ and $v_3 v_4 v_5 v_7$. We find $\beta \in \{1, 2, 7\}$ and $\beta \in \{3, 4, \alpha\}$. So $\beta = \alpha$ and $\alpha, \beta \in \{1, 2, 7\}$. Now examine cycles $v_2 v_3 v_4 v_5$ and $v_1 v_2 v_5 v_7$, this shows $\gamma \in \{2, 3, 4\}$ and $\gamma \in \{\alpha, 7, 1\}$, hence $\gamma = \alpha$ and $\alpha, \gamma \in \{2, 3, 4\}$. Since $\alpha = \beta$ and $\alpha = \gamma$, $\alpha, \beta, \gamma \in \{1, 2, 7\} \cap \{2, 3, 4\} = \{2\}$. This leaves us with cycle $v_2 v_5 v_7 v_3$ colored with only the color 2. That is, we have a monochromatic $C_4 \subseteq K_n$. $\square$

Lemma 3.18: *Assume $n \geq m \geq 9$ and m is odd. Let K_n be edge colored and suppose some $C_m \subseteq K_n$ is rainbow. Then there is a monochromatic or rainbow $C_4 \subseteq K_n$.*

Proof. Let $C_m \subseteq K_n$ be m -edge colored numerically one through m . Suppose there is no rainbow $C_4 \subseteq K_n$.

Assume $m \geq 9$. Drop a chord on C_m so that one new cycle is C_4 . Since by our assumption C_4 is not rainbow. So C_{m-2} must be rainbow. Continue dropping chords on C_{m-2}, C_{m-4} , and so forth until we have a rainbow C_7 . By Lemma 3.17, we have a monochromatic C_4 . $\square$

Lemma 3.19: *Assume $n \geq m \geq 6$. If K_n is edge colored and there is a rainbow $C_m \subseteq K_n$ then there is either a monochromatic or rainbow $C_4 \subseteq K_n$.*

Lemma 3.20: *Assume $n > 6$. Color the edges of K_n . Let $B_\ell \subseteq K_n$, where the C_3 's are connected by a path, P , with $l = |E(P)| \geq 4$. Then K_n contains either a monochromatic or a rainbow C_4 .*

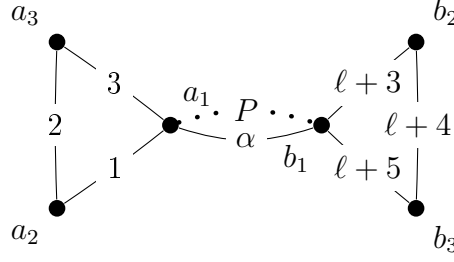


FIGURE 7. A rainbow subgraph $B_\ell \subseteq K_n$ with an additional edge, a_1b_1 , where $c(a_1b_1) = \alpha$.

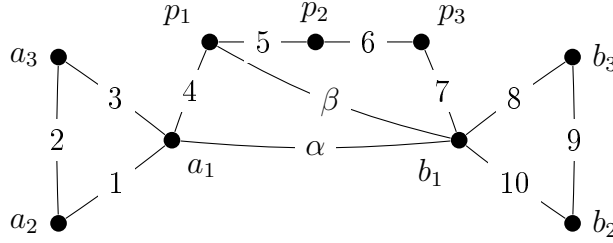


FIGURE 8. In Case i in the proof of Lemma 3.20, if $\alpha \in \{8, 9, 10\}$, then either there is a rainbow C_4 in K_n , with edges colored 5, 6, 7, β , or there is a rainbow B_1 , with edges colored 1, 2, 3, 4, β , α .

Proof. Suppose $P = a_1, p_1, p_2, \dots, b_1$. Then $a_1b_1 \in E(K_n) \setminus E(B_l)$ is an edge whose endpoints are vertices in each C_3 as depicted in Fig. 7. Let $c(a_1b_1) = \alpha$. Either $\alpha \in \{1, 2, 3, \ell + 3, \ell + 4, \ell + 5\}$ or not.

Case 1. Suppose $\alpha \notin \{1, 2, 3, \ell + 3, \ell + 4, \ell + 5\}$, then we have a rainbow $B_2 \subseteq K_n$ and can use Lemma 3.14 to conclude that K_n contains a monochromatic C_4 .

Case 2. If $\alpha \in \{1, 2, 3, \ell + 3, \ell + 4, \ell + 5\}$, then the cycle $C = a_1p_1p_2 \dots b_1a_1$ is rainbow. Either $|E(P)| = 4$ or $|E(P)| \geq 5$.

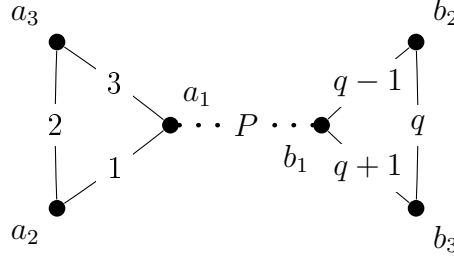
Case i. Suppose $|E(P)| = 4$. Then $C \simeq C_5$. Without loss of generality, assume that $\alpha \in \{8, 9, 10\}$ (see Fig. 8) and consider the chord p_1b_1 of C in Fig. 8. Let its color be β . If $\beta \notin \{5, 6, 7\}$ then we have a rainbow C_4 in K_n . Otherwise, if $\beta \in \{5, 6, 7\}$ then we have a rainbow B_1 , whence, by Lemma 3.10, there is a rainbow C_4 in K_n .

Case ii. Assume $|E(P)| \geq 5$. We can use our Theorem 3.19 to show that K_n contains a rainbow or monochromatic C_4 .

□

Proposition 3.21: Assume $n > 6$. Color the edges K_n . Let $B_t \subseteq K_n$ be a connected rainbow subgraph. Then K_n contains either a monochromatic or rainbow C_4 .

Proof. This follows directly from the Lemma 3.10, Lemma 3.14, Lemma 3.15, and Lemma 3.20. □

FIGURE 9. A rainbow subgraph $B_t \subseteq K_n$.

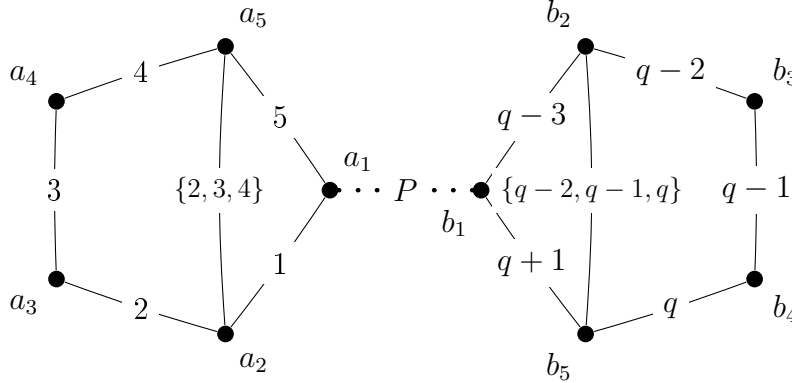
4. PROOF OF COROLLARY 1.6

Theorem 4.1: For integers $n \geq 4$ and $i \geq 1$, $n + i \notin \text{MRS}(K_n; C_4, C_4)$.

Proof. Recall Corollary 3.13. This result states that $n + i \notin \text{MRS}(K_n; C_4, C_4)$ where $n \in \{4, 5\}$, and $i \geq 1$.

Now let K_n be $(n + i)$ -edge colored where $n \geq 6$ and $i \geq 1$. Assume that such a coloring forbids rainbow and monochromatic C_4 's. By Proposition 3.7, we know K_n contains at least two cycles in the rainbow connected subgraph, say H . Let the collection of these rainbow cycles in H be denoted $rC(H)$. There are several cases we will consider where the largest rainbow cycle in $rC(H)$ is one of the following: C_m (for all $m \geq 6$), C_5 , or C_3 . There are also sub-cases we will soon address.

Case 1. Suppose the largest rainbow cycle in $rC(H)$ is C_m where $m \geq 6$. By the Theorem 3.19, we have a monochromatic or rainbow $C_4 \subseteq K_n$.

FIGURE 10. A rainbow $B_t \subseteq K_n$ among two mutually rainbow C_5 's each with a chord.

Case 2. Suppose the largest rainbow cycle in $rC(H)$ is C_5 . By Proposition 3.7, let $C_\ell \in rC(H)$ be at least one other rainbow cycle in $rC(H)$. Either $\ell = 5$ or $\ell = 3$.

Case i. Let $\ell = 5$. The two C_5 's meet at the ends of a path, P , in H . Draw two chords, as depicted in Fig. 10. By assumption there are no monochromatic or rainbow C_4 's, so we have a rainbow B_t graph. By Proposition 3.21, we have monochromatic or rainbow copies of C_4 .

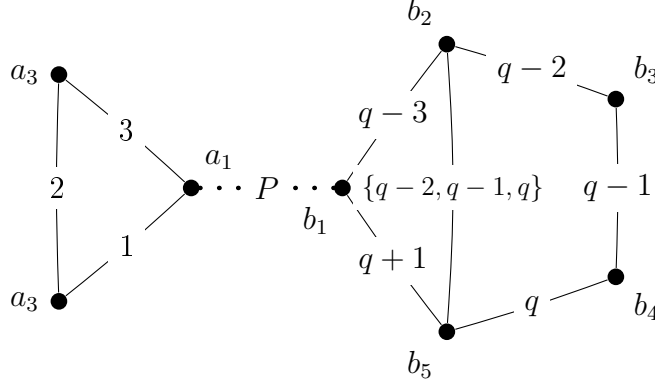


FIGURE 11. A rainbow $B_t \subseteq K_n$ among a rainbow C_3 and a rainbow C_5 with a chord.

Case ii. Let $\ell = 3$. So a C_3 and a C_5 meet at the ends of path, P , in H . Draw a chord across C_5 , as depicted in Fig. 11. By assumption there are no monochromatic or rainbow C_4 's, so we have a rainbow B_t graph. By Proposition 3.21, we have monochromatic or rainbow copies of C_4 .

Case 3. Suppose the largest rainbow cycle in $rC(H)$ is C_3 . Then we have two C_3 's that are connected by a nonempty path in H , say P . It follows that we have a B_t graph which by Proposition 3.21, we know that H has monochromatic or rainbow copies of C_4 .

Hence given a $(n+i)$ -edge colored K_n where $n > 6$ and $i \geq 1$, there is a monochromatic or rainbow $C_4 \subseteq K_n$. $\square$

It naturally follows from the previous theorems, in particular Theorem 1.7, that Corollary 1.6 holds.

Corollary 1.6 (Axenovich and Choi [2]): *For every integer $n \geq 4$,*

$$\max \text{MRS}(K_n; C_4, C_4) = n.$$

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